

Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the **Credit Card Approval Prediction** system can be scaled to support a larger number of users, process increased application data efficiently, and incorporate advanced machine learning features in the future. It also presents the team's roadmap for improving the application's performance, security, accuracy, and usability through future enhancements such as cloud deployment, database integration, explainable AI, and real-time prediction capabilities.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset size may affect prediction accuracy.	Medium	High
2	The system predicts only approval or rejection without detailed reasons.	Medium	Medium
3	The application is designed for single-user access and may not support heavy traffic.	Low	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single or few users at a time.	Deploy on cloud (AWS/Render/Heroku) with load balancing
2	Data Storage	Uses local CSV dataset.	Integrate MySQL or PostgreSQL database for large datasets.
3	Performance	Uses a trained ML model locally.	Optimize model, caching, and faster preprocessing techniques.

4	Security	Basic Flask application without authentication.	Add user login, HTTPS, and secure data encryption.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve model accuracy using larger datasets.	1 Month	More accurate predictions.
Phase 2	Deploy the application on a cloud platform.	1 Month	Better accessibility and scalability.
Phase 3	Add user authentication and prediction history.	2 Month	Improved security and user experience.
Phase 4	Integrate explainable AI to provide reasons for approval/rejection.	2 Month	Greater transparency and user trust.